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Excerpt
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THE TWENTY-NINTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

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A-1. Prove

$$\frac{22}{7} - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx.$$

A-2. Given integers a, b, e, c, d, and f with $ad \neq bc$, and given a real number $\epsilon > 0$, show that there exist rational numbers r and s for which

$$0 < |ra + sb - e| < \epsilon,$$
$$0 < |rc + sd - f| < \epsilon.$$

A-3. Prove that a list can be made of all the subsets of a finite set in such a way that (i) the empty set is first in the list, (ii) each subset occurs exactly once, (iii) each subset in the list is obtained either by adding one element to the preceding subset or by deleting one element of the preceding subset.

A-4. Given n points on the sphere $\{(x, y, z): x^2+y^2+z^2=1\}$, demonstrate that the sum of the squares of the distances between them does not exceed n^2.

A-5. Let V be the collection of all quadratic polynomials P with real coefficients such that $|P(x)| \leqq 1$ for all x on the closed interval $[0, 1]$. Determine

$$\sup\{\,|P'(0)| : P \in V\}.$$

A-6. Determine all polynomials of the form $\sum_0^n a_i x^{n-i}$ with $a_i = \pm 1\,(0 \leqq i \leqq n,\ 1 \leqq n < \infty)$ such that each has only real zeros.

B-1. The temperatures in Chicago and Detroit are x° and y°, respectively. These temperatures are not assumed to be independent; namely, we are given:
(i) $P(x^\circ = 70^\circ)$, the probability that the temperature in Chicago is 70°,
(ii) $P(y^\circ = 70^\circ)$, and
(iii) $P(\max(x^\circ, y^\circ) = 70^\circ)$.
Determine $P(\min(x^\circ, y^\circ) = 70^\circ)$.

B-2. A is a subset of a finite group G (with group operation called multiplication), and A contains more than one half of the elements of G. Prove that each element of G is the product of two elements of A.

B-3. Assume that a 60° angle cannot be trisected with ruler and compass alone. Prove that if n is a positive multiple of 3, then no angle of $360/n$ degrees can be trisected with ruler and compass alone.

B-4. Show that if f is real-valued and continuous on $(-\infty, \infty)$ and $\int_{-\infty}^{\infty} f(x)dx$ exists, then

$$\int_{-\infty}^{\infty} f\left(x - \frac{1}{x}\right) dx = \int_{-\infty}^{\infty} f(x)dx.$$

B-5. Let p be a prime number. Let J be the set of all 2×2 matrices $\binom{ab}{cd}$ whose entries are chosen from $\{0, 1, 2, \cdots, p-1\}$ and satisfy the conditions $a+d \equiv 1 \pmod p$, $ad - bc \equiv 0 \pmod p$.

Determine how many members J has.

B-6. A set of real numbers is called compact if it is closed and bounded. Show that there does not exist a sequence $\{K_n\}_{n=0}^{\infty}$ of compact sets of rational numbers such that each compact set of rationals is contained in at least one K_n.